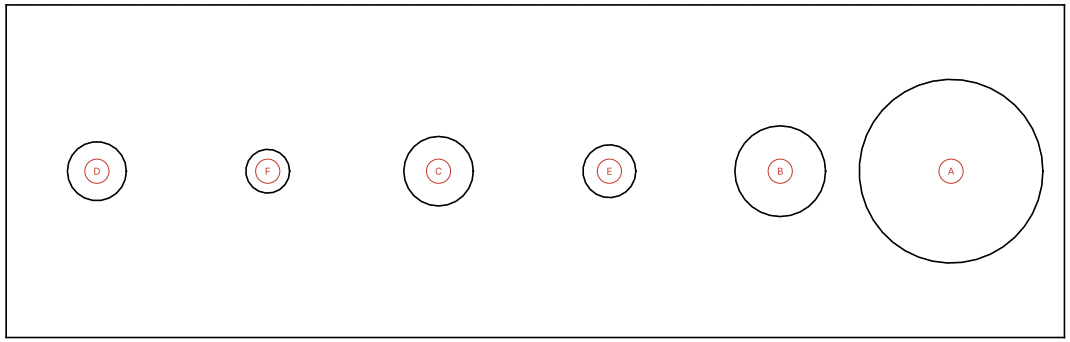
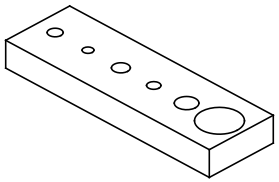
CA-900 Tolerance coupon

NOTES

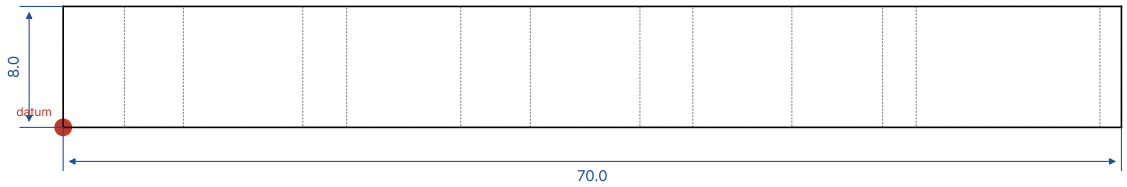
1. Hole sizes are the printed sizes from the PETG tolerance table (backend/cad/model.py TOL).
   Print CA-900 first and adjust TOL if a fit is off.
2. Screws thread straight into the printed pilots (HF kit #67675, no nuts).
3. Datum = the part's bounding-box corner (min X, Y, Z), lower left of the front and top views.
   Hole centres: (X, Y) for Z holes, (X, Z) for Y holes, (Y, Z) for X holes. ×k = k holes stacked
   along the axis; +n = n more positions (see the STEP file).
4. Shown in its assembly orientation; the STEP file defines all other geometry.

TOP VIEW

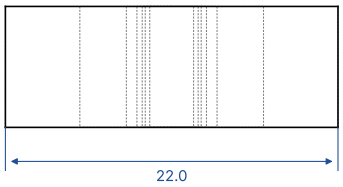




ISOMETRIC (not to scale)



FRONT VIEW



RIGHT VIEW

HOLE TABLE (letters mark hole centres on the view along that axis)

	Ø (mm)	×	axis	depth	centres from datum (mm)	fit / feature
A	12.15	1	Z	8.0	(62.5, 11.0)	LM6UU press fit
B	6.00	1	Z	8.0	(51.2, 11.0)	6 mm rod press fit
C	4.60	1	Z	8.0	(28.6, 11.0)	#8-32 clearance
D	3.90	1	Z	8.0	(6.0, 11.0)	#6-32 clearance / running
E	3.50	1	Z	8.0	(39.9, 11.0)	#8-32 pilot
F	2.90	1	Z	8.0	(17.3, 11.0)	#6-32 pilot (self-threading)

PART NO.	**CA-900**
TITLE	**Tolerance coupon**
MATERIAL	PETG printed, 0.4 mm nozzle
MASS	8.2 g (CAD, 60 % fill)
QTY	1 per robot
SCALE	2:1
UNITS	mm · general tolerance ±0.2
SOURCE	cad/step/parts/tolerance_coupon.step (from backend/cad/model.py)
PROJECTION	Third angle ⊕◁
SHEET	13 of 13
DRAWN	ME 2331 Project 2 team 2026-11-02